Role of the mutant structure (7O58), why is it important?

Structure 7O58 shows the T237M variant as a homodimer bound to the activator G16, solved by X-ray crystallography at 1.97 Å resolution. Methionine is modelled at position 237 in both chains. The structures of this variant, with and without the activator, are nearly identical to the corresponding wild-type structures (Briso-Montiano 2022).

Compared with the wild-type structure 7O1B, which was prepared in the same way, the mutant therefore keeps the same overall fold and the same dimeric assembly, and it still binds the activator. The relevant difference is local. In the wild-type protein, the side chain of Thr237 forms a hydrogen bond with Thr16, in a loop that coordinates a magnesium ion. Methionine has no hydroxyl group and cannot form this bond.

Our reading is that T237M does not harm the protein by changing its shape. The folded enzyme looks much like the normal one, but it seems to be less stable, so fewer copies end up correctly folded and active. This is also why variants like this one are good candidates for pharmacochaperones, small molecules that help a protein hold on to its folded form.